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# UAV sound source distance estimation based on ResNet-CBAM-GRU model

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***Abstract*—In recent years, the rapid development of unmanned aerial vehicle (UAV) technology has enabled its widespread application in various fields. However, with the sharp increase in the number of UAVs, issues such as privacy concerns and airspace security risks caused by their abuse have become increasingly prominent. In the interception of illegal UAV, distance estimation is also a key aspect. In this paper, a ResNet-CBAM-GRU model is proposed for estimating the distance of a UAV sound source. The model uses single-channel audio data as input, employs a residual neural network (ResNet) as the main architecture, combines the convolutional block attention module (CBAM) to focus on key acoustic features, and utilizes a gated recurrent unit (GRU) to capture temporal dynamics, thereby improving distance estimation accuracy. The experimental results show that the mean absolute error (MAE) of this model in the distance estimation of unmanned aerial vehicle sound sources in outdoor environments is 0.5275, and the mean relative error (MRE) is 0.0455. These results verify the effectiveness of the model in estimating the distance of unmanned aerial vehicle sound sources.**

***Keywords-component; UAV sound source distance estimation; Single-channel audio; Convolutional block attention module.***

## I. INTRODUCTION

With the continuous advancement of the low-altitude economy and unmanned aerial vehicle (UAV) technology, UAVs have become indispensable tools in several key fields, such as agricultural monitoring and emergency rescue[1-3]. However, as the number of drones has sharply increased, so too has the potential risk of their abuse. Such misuse poses a threat to public safety and social stability. Therefore, it has become particularly urgent to develop and deploy effective countermeasures against unmanned aerial vehicles[4-5].

The core of drone countermeasure technology is to identify and intercept illegal drones. Precise interception requires first locking onto the target position. Currently, in research on sound source localization of UAVs, sound [6] is identified and located by using microphones to capture the unique noise generated by UAV rotors. Reference [7] addresses the privacy and security issues associated with the abuse of amateur UAVs by designing a monitoring system based on a double tetrahedral acoustic array. By utilizing the SVM detection fusion algorithm and the Bayesian and Gaussian mixture model TDOA algorithm, the system achieves real-time, high-precision positioning of UAVs. Reference [8] aimed at the problems of weak feature extraction and poor positioning in traditional acoustic positioning under low signal-to-noise ratio. It proposed deep learning-based positioning technology and utilized a delay estimation neural network DOA algorithm to improve the performance of traditional algorithms in terms of resolution and robustness under low signal-to-noise ratio conditions.

With the increasingly in-depth application of neural networks in the task of sound source distance estimation, remarkable achievements have been made in related research. Reference [9] proposed a method based on transfer learning and a residual neural network (ResNet). By pre-training with a large amount of acoustic data from the selected sea area and fine-tuning with a small amount of data from the target sea area, the distance estimation problem under few-sample conditions in shallow water environments was solved. The results show that this method can achieve more accurate distance estimation in scenarios with limited samples. Reference [10] constructed a sound source localization model (CNN-R) integrating convolutional neural networks and regression models, and utilized the short-time Fourier transform (STFT) to extract the binaural phase difference (IPD) features of sound signals for estimating the angle and distance of the sound source. Experiments show that the average accuracy rates of angle and distance estimation in the simulated scenarios are 98.96% and 98.31%, respectively, and can reach 99.85% and 99.38% in the real environment. Reference [11] proposed a convolutional recurrent neural network (CRNN) with an attention mechanism to achieve continuous distance estimation of audio signals for speaker distance estimation in a single-channel audio environment. In tests on multiple datasets, the absolute error in the noise-free synthetic scene is 0.11 m, that in the mixed scene is approximately 1.30 m, and that in the real scene is approximately 0.50 m.



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The above-mentioned scholars have all proposed reasonable schemes in the research on unmanned aerial vehicle sound source localization and sound source distance estimation. They have demonstrated excellent performance in actual environmental tests, making significant contributions to technological progress in related fields. This paper proposes a neural network model, ResNet-CBAM-GRU, based on single-channel audio signal input, to address the problem of sound source estimation for UAV. The model uses ResNet network [12] as the main architecture, embeds the Convolution Block Attention Module (CBAM)[13], and strengthens the extraction ability of key features through channel attention mechanism and spatial attention mechanism. At the same time, to effectively capture the long-term and short-term dependencies in the audio sequence, the gated recurrent unit (GRU) module is introduced to optimize temporal feature processing. Finally, the effectiveness and superior performance of the proposed model in sound source distance estimation are verified by the UAV sound data set.

## II. PROPOSED APPROACH

### *A. The model structure of UAV sound source distance estimation*

In this study, a ResNet-CBAM-GRU network model for UAV sound source distance estimation using single-channel audio is designed. The overall structure of the model is shown in Figure 1.

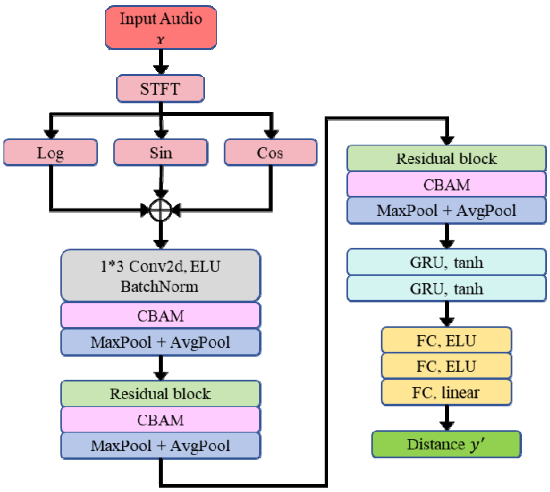


Figure 1. Structure of UAV sound source distance estimation model.

The model is mainly composed of four parts: the first part is the acoustic feature input section, which converts the single-channel UAV audio signal into the frequency domain through STFT, and extracts the amplitude and phase information of the signal by applying logarithm ( $log$ ), sine ( $sin$ ), and cosine ( $cos$ ) operations, respectively. After fusion, this information is used as the model input. Secondly, for the feature extraction part, a three-level convolutional structure is adopted, consisting of an initial convolutional layer and two residual blocks. The CBAM attention mechanism is embedded after the initial convolutional layer and the two residual blocks. The output of each level module is dimensionally reduced by combining max pooling and average pooling. The next one is the loop layer composed of two gated recurrent units (GRUs), which can effectively capture both long-term and short-term dependencies in the audio sequence during audio processing tasks. Finally, there are three fully connected layers that map the output features from the loop layer to the distance of the UAV's sound source.

The initial convolution layer uses a 1×3 convolution kernel and an ELU activation function. The expression of ELU is:

$$ELU = \begin{cases} x & x > 0 \\ \gamma\left(e^{x} - 1\right) & x \le 0 \end{cases} \tag{1}$$

Where $x$ denotes the input feature; $\gamma$ is a hyperparameter, which adjusts the mean value of the output to near 0.

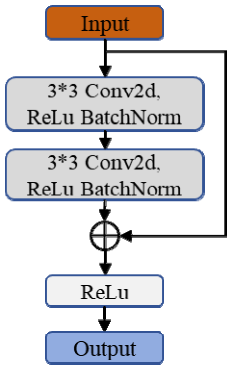


Figure 2. Residual block structure.

The structure of the residual block is shown in Figure 2. Unlike the initial convolutional layer, the convolutional layers inside the residual block use 3×3 convolutional kernels and ReLU activation functions. The expression of ReLU is:

$$ReLU = \begin{cases} x & x \ge 0 \\ 0 & x < 0 \end{cases} \tag{2}$$

### *B. Convolution Block Attention Module (CBAM) Attention Module*

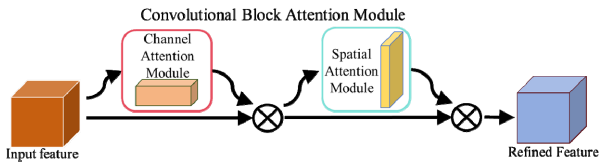


Figure 3. CBAM attention module.

CBAM is a lightweight, end-to-end trainable feature recalibration module that enhances the feature representation capability and discriminative power of Convolutional Neural Networks (CNNs) without significantly increasing computational complexity or the number of parameters. Its structure is shown in Figure 3. The CBAM combines channel attention (CAM) and spatial attention (SAM) attention mechanisms. By giving an intermediate feature matrix, the



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module can infer the attention matrix along the two independent dimensions of channel and space, and then multiply it with the input feature matrix to achieve adaptive feature optimization.

The Channel Attention Mechanism (CAM) compresses the feature matrix in the spatial dimension while retaining the channel feature information. Firstly, the input feature matrices are subjected to global average pooling ( $GAP$ ) and global max pooling ( $GMP$ ), respectively. Subsequently, these two vectors are fed into a shared multi-layer perceptron ( $MLP$ ) for processing to generate the corresponding channel weights. After summing these two weight vectors and normalizing the result through the Sigmoid function, the final channel attention feature map is obtained. Its mathematical expression can be expressed as:

$$M_c(\mathbf{X}) = \sigma_S\left(MLP\left(GAP(\mathbf{X})\right) + MLP\left(GMP(\mathbf{X})\right)\right) \quad (3)$$

Where $\mathbf{X}$ represents the input feature matrix and $\sigma_S$ represents the Sigmoid function.

The Spatial Attention Mechanism (SAM) compresses the feature matrix along the channel dimension while retaining the spatial feature information. First, average pooling ( $AvgPool$ ) and max pooling ( $MaxPool$ ) are performed on the input feature matrix. Then, the results of these operations are concatenated along the channel dimension and passed through a convolutional layer. Finally, the Sigmoid function is used for normalization to generate the spatial attention weights. Its mathematical expression can be written as:

$$M_S(\mathbf{X}) = \sigma_S\left(f^{7\times7}\left(\left[AvgPool(\mathbf{X}); MaxPool(\mathbf{X})\right]\right)\right) \quad (4)$$

### C. Gate Recurrent Unit

In the network structure designed in this paper, there are two gated recurrent unit (GRU) layers, whose activation function is tanh. This activation function is used to handle the feature matrix output from the feature extraction part. These recurrent layers undertake the core task of temporal modeling and adaptively learn the long-term dependencies in acoustic features through the gating mechanism. In the distance estimation of UAV sound sources, the influence of instantaneous interferences such as wind noise and sudden noise sources can be reduced, thereby enhancing the extraction of stable acoustic features.

GRU is similar to the Long Short-Term Memory (LSTM) network[14], which aims to overcome the vanishing gradient problem in recurrent neural networks (RNNs), thereby enabling more effective handling of long-term dependencies in sequential data. Compared with LSTM, the design of GRU is more concise. It combines the forget gate and the input gate into a single update gate. This reduces the number of model parameters, improves computational efficiency, and at the same time retains the ability to capture long-term dependencies.

## III. Experimental Setup and Results

All experiments in this paper were conducted in the Python programming environment, version 3.8. The network was built using the PyTorch library, and the PyTorch Lightning framework was employed for model training and testing.

Some of the equipment and the environment involved in the experiment are shown in Figure 4. Among them, Figure 4. (a) shows the DJI Air 2S UAV used for sound recording; Figure 4. (b) shows the MPA201 microphone used for sound recording, which has a frequency response range of 10 Hz to 20 kHz. Figure 4. (c) shows the Cube4Nano sound card, which has a frequency response range of 20 Hz to 20 kHz. Figure 4. (d) shows an outdoor open recording site where various environmental noises exist.

**(a) DJI Air 2S** **(b) MPA201 microphone**

**(c) Cube4Nano sound card** **(d) Experimental scene**

Figure 4. Experimental equipment and scene.

### A. Datasets and Evaluation Metrics

In order to verify the effectiveness of the established model in UAV sound source distance estimation, this paper uses the UAV sound dataset collected in the laboratory to test the performance of UAV sound source distance estimation. The audio in the drone sound dataset was recorded using the DJI Air 2S drone in an outdoor environment to study how the drone's sound characteristics vary with distance. The audio coverage distance range in the dataset is 5–21 m, and the sampling rate used for audio recording is 48 kHz. The original audio file obtained from the recording was cropped into a 3-second audio clip as a sample. The sample distribution at various distances in the dataset is shown in Figure 5. , and the average distance of all samples is 12.94 m.



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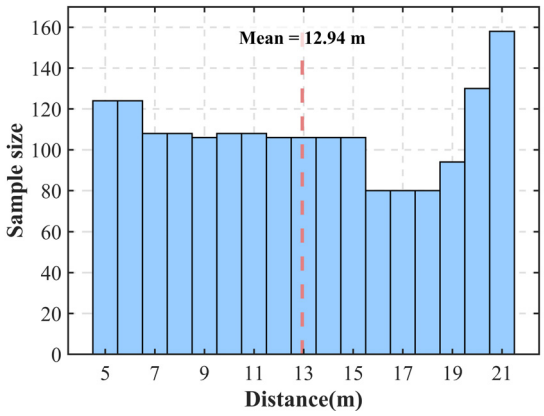


Figure 5. Sample distribution at different distances.

In this paper, to evaluate the performance of the model, the Mean Absolute Error (MAE) is adopted to describe the overall performance of the model on the test set. MAE can visually reflect the mean absolute deviation and is robust to outliers. The mathematical expression of the average absolute value error is:

$$MAE\left(y,y'\right)=\frac{1}{m}\sum_{i=1}^{m}\left|y_i-y_i'\right| \tag{5}$$

Where $m$ is the sample size of the test set, $y$ represents the true value, and $y'$ represents the estimated value. The relative error of the model at different sound source distances is quantified by calculating the MAE within various distance ranges, and then the performance of the model is evaluated.

At the same time, in order to further evaluate the performance of the model, this paper introduces the mean relative error (MRE) and the root mean square error (RMSE). MRE is used to quantify the average deviation ratio of the model, and RMSE is used to measure the average deviation degree of the model. Their mathematical expressions are:

$$MRE\left(y,y'\right)=\frac{1}{m}\sum_{i=1}^{m}\left|\frac{y_i-y_i'}{y_i}\right| \tag{6}$$

$$RMSE\left(y,y'\right)=\sqrt{\frac{\sum_{i=1}^{m}\left(y_i-y_i'\right)^2}{m}} \tag{7}$$

MRE has the characteristic of scale invariance, which can effectively eliminate the influence of dimensional and scale differences in the original data, making it an ideal tool for evaluating model accuracy across datasets of different dimensions and scales. Meanwhile, its calculation results can directly reflect the average estimation deviation of the model. The RMSE reflects the average level of distance estimation errors and indicates the influence of extreme errors.

### *B. Verification Results of the UAV sound dataset*

In this test, to verify the effectiveness of the model in the task of estimating the distance of unmanned aerial vehicle sound sources, the improved ResNet-CBAM-GRU, ResNet, and CRNN network models were compared. Meanwhile, to evaluate the role of the GRU module, the sound source distance estimation accuracy of each model was tested in both cases: with GRU and without GRU. Meanwhile, to understand the fluctuation of the model's error in the task of estimating the distance of UAV sound sources, the standard deviations of MAE and MRE were calculated to quantify the degree of dispersion of the error distribution.

TABLE I. ESTIMATION RESULTS OF UNMANNED AERIAL VEHICLE DISTANCE.

| Model | GRUs | MAE | MRE | RMSE |
|---|---|---|---|---|
| ResNet-CBAM-GRU | 0 | 0.9755±0.8489 | 0.0819±0.0704 | 1.2903 |
| | 2 | 0.5275±0.4650 | 0.0455±0.0387 | 0.7027 |
| ResNet | 0 | 0.8327±0.7272 | 0.0696±0.0593 | 1.1049 |
| | 2 | 0.6308±0.5520 | 0.0517±0.0417 | 0.8377 |
| CRNN | 0 | 1.6611±1.3452 | 0.1498±0.1400 | 2.1363 |
| | 2 | 0.7506±0.5925 | 0.0663±0.0549 | 0.9558 |

The verification results of each model on the UAV sound source dataset are shown in TABLE I. The results show that, compared with several other model structures, the MAE of the proposed model in the distance estimation of unmanned aerial vehicle sound sources is 0.5275, with a standard deviation of 0.4650, demonstrating the high accuracy of this model's distance estimation. The MRE is 0.0455, with a standard deviation of 0.0387, and the RMSE is 0.7027, further verifying that the overall estimation is closer to the true value. In the experimental results, the model with the residual block introduced showed significantly higher estimation accuracy than the conventional convolutional neural network in the absence of the GRU module, verifying that the residual block can effectively enhance the model's ability to extract the acoustic features of UAV. Furthermore, when the GRU module was added to all models, the distance estimation accuracy was significantly improved compared with the models without the GRU module, fully demonstrating the key role of the GRU module in enhancing the model's ability to process audio signals. The experimental results effectively verify the validity of the model proposed in this paper in the distance estimation of UAV sound sources.

Figure 6. shows the scatter plots of the estimation results of the ResNet-CBAM-GRU model proposed in this paper, the ResNet-CBAM model without the GRU module, the ResNet model with the GRU module added, and the CRNN model on the test set. The determination coefficients ( $R^2$ ) of each model were 0.9796, 0.9312, 0.9710, and 0.9623, respectively. The $R^2$ values of all models are higher than 0.9, indicating that they are all effective in the distance estimation of UAV sound sources. Among them, the determination coefficient of the proposed model is greater than those of the other models, verifying its higher accuracy in estimating UAV sound source distances.



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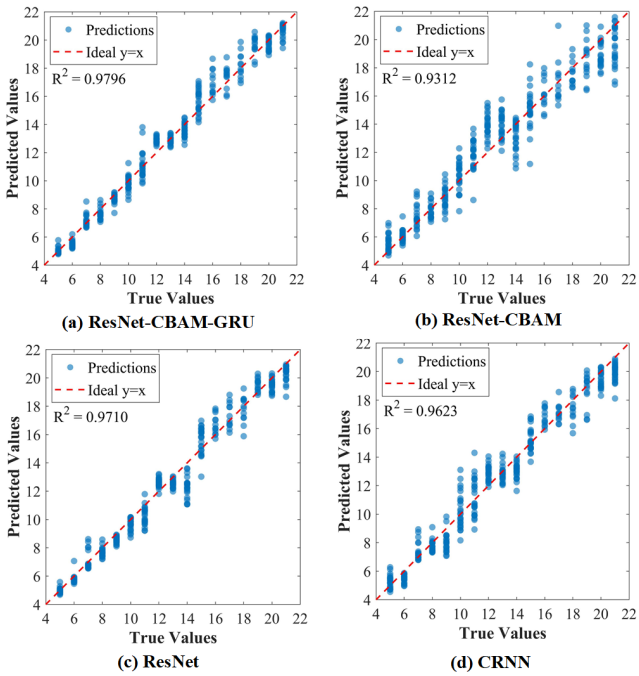


Figure 6. The verification results of each model on the test set.

## IV. Conclusions and Outlook

In this paper, a ResNet-CBAM-GRU model that takes single-channel audio data as input is proposed for UAV sound source distance estimation. This model takes ResNet as the backbone network and introduces the CBAM attention mechanism, enabling the network to better focus on the features related to distance in the unmanned aerial vehicle (UAV) sound signal. Meanwhile, the GRU model is added to enhance the model's ability to capture long-term dependencies in time series. The experimental results show that the mean absolute error (MAE) of the proposed model in the estimation of UAV sound source distance can reach 0.5275; the standard deviation used to describe the error fluctuation is 0.4650; the mean relative error (MRE) can reach 0.0455; and the standard deviation is 0.0387. The effectiveness of the model in UAV sound source distance estimation is verified. However, there are still some shortcomings in this study. The existing datasets lack sound source distance data under the motion state of the UAV, which leads to the lack of distance estimation ability of the model in dynamic scenes. In subsequent research, the relevant data will be further supplemented, and the model structure will be improved to enhance the model's generalization ability in estimating UAV sound source distance in dynamic scenes.

## Acknowledgment

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